

Measures of Center

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In summary:

$$\text{mean } \bar{x} = 68.9$$

$$\text{median } \tilde{x} = 73.0$$

$$\text{mode } M = 73$$

which is Better?

Given a Quantitative (numeric) set of Data,
one can ask the question: What is a typical value of the data?

For Example:

Suppose these are exam scores

{7, 64, 67, 70, 72, 73, 73, 73, 76, 82, 85, 85}

This question can be answered as follows:

I. The Mean - the average in which you add all the values and divide by the number of values.

$$\mu = 68.9 \quad \bar{x} = 68.9$$

population μ
("mean")

sample $\bar{x}$
("x-bar")

II. The Median - this is the center value in a set of sorted numbers. If an odd number of data, take the exact middle; if an even number of data, take midpoint of the two middle values.

$$\tilde{x} = 73.0$$

"x-kidde"

* NOTICE
ROUNDING

III. The Mode - this is the most frequently occurring value; sometimes there might be two modes and this is called bi-modal. If more than two modes, it is then called multi-modal; however, that has no practical value.

$$M = 73$$

no rounding

Note: throughout this textbook, measures of center are referred to as ~~Typical Values~~; however, that is NOT the terminology used in Statistics. These values are called Measures of Center.

Measures of Center tell us approximately where the center of the data is.

I. The Mean

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scratch

$$59/4 = 14.75$$

FINDING THE MEAN

A **typical value** is an ^{datum} amount that represents the values in a set of numbers. A typical value must fall within the series of numbers it represents. There are three common typical values in a set of numbers: **mean**, **median**, and **mode**.
1. 2. 3.

When most people use the word **average**, they are talking about the **mean**. For a set of values that are similar to each other, the mean is what you want to know when you ask, "About how much . . . ?"

To find the mean, add the values in a set of data, and then divide the sum by the number of values in the set.

FINDING THE MEAN

Example Ella scored 14, 18, 11, and 16 points in her last 4 basketball games. What is her scoring average?

Step 1

Find the mean. Add the values in the set.

$$14 + 18 + 11 + 16 = 59$$

Ella's scoring average is **14.75 points**.

Step 2

Divide the sum by the number of values in the set.

$$\frac{59}{4} = 14\frac{3}{4} = 14.75$$

The mean is not necessarily equal to any number in the set. However, it must fall within the range of the numbers in the set.

$$\{11, 14, 16, 18\} \quad \mu = 14.8$$

Population, Not a Random Sample

Examples of the Mean

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When calculating the mean (average), always write one more decimal place for the mean. For example, if the data are whole numbers, write the mean with one decimal point. The only time you would break this rule is if the extra decimal makes no sense; e.g., a mean of \$1.583 makes no sense! Also, when rounding, if 5 or more, round up; if less than 5, round down.

ASSUME THESE ARE ALL SAMPLES.

Find the mean for each set of numbers.

1. 35, 49, 31, 25

$$\bar{x} = 35.0$$

\$1.25, \$2.08, \$0.97, \$3.12, \$1.48

$$\bar{x} = \$1.78$$

common sense, rounding *

5, 4, 12

$$\bar{x} = 7.0$$

2. 174, 12, 398, 214, 45

$$\bar{x} = 168.6$$

6, 3, 15, 9, 4

$$\bar{x} = 7.4$$

\$35, \$12, \$97, \$6, \$51, \$90

$$\bar{x} = 48.50$$
 *

3. \$215, \$154, \$192

$$\bar{x} = \$181$$
 *

87, 96, 95, 100, 75, 82, 88

$$\bar{x} = 89.0$$

32.1, 15.75, 17.2, 18.35, 28.4

$$\bar{x} = 22.36$$

4. 5.6, 5.1, 6.4

$$\bar{x} = 5.70$$

\$240, \$198, \$264, \$249, \$220

$$\bar{x} = \$234.20$$
 *

1, 7, 13, 7, 10

$$\bar{x} = 7.6$$

Solve.

5. In July, DeShaun's weekly grocery bills were \$84, \$98, \$72, and \$60. What is his mean weekly grocery bill for the month?

NOT A RANDOM SAMPLE

$$\mu = \$78.50$$

Pop: all weeks in a particular month

6. Miranda received scores of 83, 78, 85, 82, and 77 on 5 tests. What is her average score on the 5 tests?

Not sure if these are all her tests μ or just a sample $\bar{x}$

$$81.0$$

Examples of the Mean, cont'd...

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If you know the mean, you can use algebra to find a missing value. You can also use the mean to draw conclusions and make predictions.

There are called:

CONFIDENCE INTERVALS
& HYPOTHESIS TESTING

USING THE MEAN

Example 1 Sean budgets \$400 a month for food. In 5 months, he spent \$415, \$380, \$489, \$392, and \$404. In order to average \$400 per month, how much should he spend in the sixth month for food?

Step 1

Write an equation to represent the mean. Let x represent the amount spent in the sixth month.

$$\frac{415 + 380 + 489 + 392 + 404 + x}{6} = 400$$

Sean should spend **\$320** in the sixth month to average \$400 per month.

Example 2 Gina is paid weekly based on the amount of sales she makes. Her first paychecks are for the following amounts: \$346, \$358, \$320, and \$336. Based on the mean for the first 4 weeks, about how much will Gina make in a year?

Step 1

Find the mean of the first 4 weeks.

$$\frac{346 + 358 + 320 + 336}{4} = 340$$

$$\mu = \$340$$

Gina will make about **\$17,680** in a year.

Step 2

Solve for x .

$$\frac{415 + 380 + 489 + 392 + 404 + x}{6} = 400$$

$$\frac{2,080 + x}{6} = 400$$

$$2,080 + x = 2,400$$

$$x = 320$$

special note

if there are n data

then there are $n-1$

Degrees of Freedom

$\therefore n=6$ means
 $df=5$ } there is only
one possible
answer for x .

Step 2

Gina will typically make \$340 a week. There are 52 weeks in a year. Multiply the mean by 52.

$$\$340 \times 52 = \$17,680$$

yearly income $\rightarrow \$17,680$

Examples of the Mean, cont'd...2...

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Solve as directed.

7. Jena plans to run an average of 6 miles a day. For the first 6 days, she runs 3, 8, 8, 0, 4, and 5 miles. How many miles should she run on the seventh day to meet her goal?

$$\begin{aligned}\mu &= 6.0 \\ \frac{3+8+8+0+4+5+x}{7} &= 6 \\ x &= 14 \text{ miles}\end{aligned}$$

8. In the first 8 weeks of school, students at Pine Elementary read the following numbers of books.

80 85 92 78 80 81 90 96

Based on the average of the first 8 weeks, how many books will they read in a 36-week school year?

$$\begin{aligned}\mu &= 85.25 \\ &= 85.3 \text{ per wk} \\ 36 \text{ wks} (85.3) &= 3070.8 \\ &= 3071\end{aligned}$$

9. Jared set a goal to sell 300 alarm systems in a 60-day period. At the end of the 60 days, he had averaged 4 alarms per day. Did he meet his goal?

$$60 \times 4 = 240$$

NO.

10. **Explain** Carmel received test scores of 77, 81, 95, and 91. She has 1 more test to take. In order to get an A in the class, she needs an average of 90 on the 5 tests. If the highest score Carmel can earn on the last test is 100, is it possible for her to get an A in the class? Explain your thinking.

$$\frac{77+81+95+91+x}{5} = 90$$

$$x = 450 - 344$$

$$x = 106$$

Not possible!!

II. The Median

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MEDIAN

There are other typical values that might be more representative of a set of data. The median is the middle value of a set of data. To find the median, sort the data from least to greatest. For an odd number of values, the middle number is the median. For an even number of values, the median is the mean of the two middle numbers.

FINDING THE MEDIAN

$\{10, 30, 35, 40, 140\}$

Example Angela has helped 5 dogs onto the examination table. Their weights are 40 lb, 10 lb, 140 lb, 35 lb, and 30 lb. What is the median weight of the dogs?

$\bar{x} = 35.0$ lb

Step 1

Sort the numbers numerically from least to greatest.

10 lb 30 lb 35 lb 40 lb 140 lb.

The median weight is **35 lb**.

Step 2

Choose the middle number from the set. (If there is an even number of values, find the mean of the two middle values.)

Find the median.

1. $\$12, \$75, \$4, \$23, \$51$

$\{4, 12, 23, 51, 75\}$
 $\bar{x} = 23$

53, 22, 12, 18, 50

$\{12, 18, 22, 50, 53\}$
 $\bar{x} = 22.0$

13.4, 8.6, 4.1, 80.9

$\{4.1, 8.6, 13.4, 80.9\}$
 $\bar{x} = (8.6 + 13.4) / 2$
 $= 11.00$

2. $0.8, 1.4, 2.3, 0.6, 1.8$

$\bar{x} = 1.40$

$62, 78, 73, 45, 67, 90$

$\bar{x} = \frac{73 + 67}{2}$

$\bar{x} = 70.0$

$\$457, \$375, \$223, \480

$\bar{x} = (457 + 375) / 2$

$\bar{x} = \$416$ * money - common sense rounding

check
 $\{0.6, 0.8, 1.4, 1.8, 2.3\}$

III. The Mode

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MODE

Measure of Center

Another typical value is the ~~mode~~—the value in the set that occurs most often. (If each value occurs only once, there is no mode in the set.)

FINDING THE MODE

Example A restaurant offers these prices on complete meals: Fried Chicken—\$8.99, Steak—\$10.99, Fish—\$8.99, Shrimp—\$10.99, Lasagna—\$8.99, Hamburger—\$7.99. What is the mode for these prices?

$$m = \$8.99$$

Count how many times each value occurs.

\$7.99: 1 time \$8.99: 3 times \$10.99: 2 times

Because it occurs the most (3 times), \$8.99 is the mode.

Find the mode.

3. 35, 18, 21, 21, 18, 97, 32, 18
 $m = 18$

22°, 17°, 31°, 22°, 18°, 35°, 22°, 31°
 $m = 22^\circ$

4. \$6.50, \$5.75, \$6.00, \$5.50, \$6.75
No mode

23, 24, 26, 24, 28, 23, 25, 23, 27, 25, 24
23 3x's
24 3x's

Bimodal

$$m = 23 \text{ \& } 24$$

CONSIDER A PRIMARY SCHOOL



AGES OF
EVERYONE IN
THE SCHOOL

WHY??

Outliers

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note: data (plural)
datum (singular)

An outlier is a datum that is significantly far from the other data.

For example, consider the data set {23, 69, 73, 76, 77, 82, 85, 88, 89, 91, 94, 99}.

Which datum is a far from the others? 23

Therefore, 23 is an outlier.

CALCULATE : mean $\mu = 78.8$

Median $\tilde{x} = 83.5$

Trimmed mean $\mu = 83.9$

Comparing the Mean & the Median

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Sometimes, one or more data values are very large or very small compared to the other values in the set. These extreme values are called outliers. Outliers don't affect the median as much as they affect the mean.

*outlier at all!!
(usually)*

USING THE MEAN AND MEDIAN WITH OUTLIERS

Example A jewelry store has 5 different rings on sale for \$27, \$26, \$33, \$31, and \$108.

Your common sense tells you that \$30 is a typical price for a ring. Compare how the outlier of \$108 impacts the mean more than the median.

Find the trimmed mean $\mu = \$29.25$

Find the mean of the 5 rings:

$$\begin{aligned} \$27 + \$26 + \$33 + \$31 + \$108 &= \$225 \\ \$225 \div 5 &= \$45 \end{aligned}$$

$$\mu = \$45$$

The mean, \$45, is much higher than \$30.

Find the median of the 5 rings:

$$\begin{array}{ccccc} \$26 & \$27 & \$31 & \$33 & \$108 \\ \text{The middle value is } & \$31. & & & \end{array}$$

$$\tilde{x} = \$31$$

The median, \$31, is very close to \$30.

Notice in the example above that while the mean may not be a typical value, the median is.

a good measure of center

First, identify the outlier. Then find the mean and the median of the data set.

5. 13.4 kg, 14.2 kg, 3.2 kg, and 12.8 kg

Outlier: 3.2

Mean: $\mu = 10.90$

Median: $\tilde{x} = 13.10$

$$\sigma = 4.47$$

$$\begin{aligned} \mu \pm 2\sigma &= 10.90 \pm 2(4.47) \\ &= 1.96 \text{ to } 19.84 \end{aligned}$$

7. 61 in, 68 in, 44 in, 60 in, 69 in, and 64 in

Outlier: 44

Mean: $\mu = 61.0$

Median: $\tilde{x} = 62.5$

$$\sigma = 8.3$$

$$\begin{aligned} 61.0 \pm 2(8.3) \\ = 44.4 \text{ to } 77.6 \end{aligned}$$

6. \$126,000; \$74,000; \$86,000; \$85,000; and \$84,000

Outlier: \$126K

Mean: $\mu = \$91K$

Median: $\tilde{x} = \$85K$

$$\sigma = 18$$

$$\begin{aligned} \mu \pm 2\sigma &= 91 \pm 2(18) \\ &= 55 \text{ to } 127 \end{aligned}$$

8. 73, 69, 76, 107, 80

Outlier: 107

Mean: $\mu = 81.0$

Median: $\tilde{x} = 76.0$

$$\sigma = 13.5$$

$$\begin{aligned} \mu \pm 2\sigma &= 81.0 \pm 2(13.5) \\ &= 54 \text{ to } 108 \end{aligned}$$

$$= 55 \text{ to } 127$$

$$= 54 \text{ to } 108$$

Solve.

9. The following data shows the number of sales made in a month by 8 out of 9 salespeople.

45 34 31 48 37 43 32 42

The 9th employee made 66 sales in that month. By how much does the mean change when the 9th value is added?

$$\mu_8 = 39.0 \quad \bar{x}_8 = 39.5$$

$$\mu_9 = 42.0 \quad \bar{x}_9 = 42.0$$

10. **Explain** The data below gives the margin of victory for a high school football team's games in a season.

~~3~~ 7 ~~21~~ ~~14~~ ~~3~~ ~~17~~ ~~30~~ ~~3~~

Do you think the mean, median, or mode is the most reliable typical value for the data? Explain

$$\mu = 12.3$$

$$\bar{x} = 10.5$$

$$M = 3$$